

PREDICTING CUSTOMER CHURN

A DATA-DRIVEN ANALYSIS OF SUBSCRIPTION CUSTOMER BEHAVIOUR USING MACHINE LEARNING

**Project Report Submitted in fulfillment of the Requirements for the Data Science & AI Training Program by
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Data Science and AI

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DECLARATION

I declare that this project report presents the analysis and modelling work completed as part of the IDRA Data Science & AI training program. The computational work and interpretations are documented in the accompanying notebook, and external sources used in preparing the report are acknowledged.

ACKNOWLEDGEMENT

I sincerely acknowledge the India Data Research Academy and the Data Science & AI training program for providing the learning framework, project guidance and dataset resources required for this capstone. I also acknowledge the project supervisor and all instructors whose training supported the completion of this work.

ABSTRACT

This study investigates customer churn in a telecommunications subscription dataset containing 7,043 customer records and 21 variables. The objective is to identify customer characteristics associated with churn and develop a classification model capable of identifying customers at risk of leaving. Data preparation included converting TotalCharges from text to numeric format, resolving 11 blank charge records using a zero value consistent with their zero-month tenure, verifying that no duplicate records were present, and excluding the customer identifier from modelling. Exploratory analysis examined churn by contract type, internet service and payment method, while statistical analysis compared tenure, monthly charges and total charges across churn outcomes. Logistic Regression was used as a baseline and Random Forest as the final model. On a stratified 80/20 test split, the Random Forest achieved 0.763 accuracy, 0.537 precision, 0.783 recall and 0.637 F1-score, with ROC-AUC of 0.842. Contract type, tenure, service-related variables and payment method were among the strongest predictive inputs. The findings support targeted retention strategies focused particularly on month-to-month customers and early-tenure customers, while emphasizing that observed relationships are not causal evidence.

Keywords: Customer Churn; Exploratory Data Analysis; Classification; Random Forest; Predictive Analytics; Customer Retention

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1. INTRODUCTION

1.1 Background

Customer churn refers to customers ending a subscription relationship. For subscription businesses, understanding churn is important because retention efforts can be directed toward customers showing characteristics associated with higher departure risk.

1.2 Context of the Problem

The dataset represents telecommunications customers with demographic, service, contract, payment and billing variables. The practical stakeholder is a customer-retention team that wants evidence about which segments experience higher churn and a model that can prioritize potential churners.

1.3 Scope of the Study

The study covers data quality assessment, preprocessing, EDA, descriptive statistics, classification modelling and interpretation. It does not claim causal effects, customer-level intervention outcomes, or production deployment performance.

2. PROBLEM STATEMENT AND MOTIVATION

The central problem is: ****Which customer characteristics are associated with churn, and how effectively can machine learning predict whether a customer will churn?*** A reliable screening model can help retention teams allocate attention more efficiently.

3. OBJECTIVES

- Understand and validate the structure and quality of the LMS-provided churn dataset.
- Identify patterns and customer groups associated with churn.
- Prepare features using leakage-safe preprocessing.
- Build and compare classification models for churn prediction.
- Evaluate the final model using accuracy, precision, recall, F1-score and a confusion matrix.
- Translate analytical and modelling results into practical retention recommendations.

4. RESEARCH QUESTIONS

1. What are the major characteristics and quality issues in the dataset?
2. Which customer attributes and service groups are most strongly associated with churn?
3. How do tenure and monthly charges differ between churned and retained customers?
4. Which features contribute most to predictive performance?
5. How accurately can churn be predicted on unseen test data?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used for this capstone is the Project 4 Customer Churn CSV supplied in the IDRA LMS Study Material. The file is used as provided; the analysis documents transformations applied during cleaning.

5.2 Dataset Size and Structure

The raw dataset contains 7,043 observations and 21 columns. The target variable is Churn, with values Yes/No. The modelling task is binary classification.

Measure	Value
Rows	7,043
Columns	21
Target	Churn
Target classes	No / Yes
Churned customers	1,869
Overall churn rate	26.54%

5.3 Data Dictionary

Variable	Meaning	Type	Role
customerID	Unique customer identifier	Categorical	Identifier / excluded
SeniorCitizen	Senior-citizen indicator	Binary numeric	Feature
Partner / Dependents	Household characteristics	Categorical	Features
tenure	Months with the service	Numeric	Feature
PhoneService / MultipleLines	Phone-service details	Categorical	Features
InternetService	Internet service type	Categorical	Feature
OnlineSecurity / OnlineBackup / DeviceProtection	Optional service indicators	Categorical	Features
TechSupport / StreamingTV / StreamingMovies	Support and entertainment services	Categorical	Features
Contract	Contract term	Categorical	Feature
PaperlessBilling / PaymentMethod	Billing and payment characteristics	Categorical	Features
MonthlyCharges	Current monthly charge	Numeric	Feature
TotalCharges	Accumulated charges	Numeric	Feature
Churn	Whether customer left	Categorical	Target

6. DATA CLEANING

6.1 Missing Values

The raw file has no formal CSV null values, but TotalCharges contains 11 blank strings. After conversion to numeric, these become missing. All 11 affected records have tenure equal to 0, so TotalCharges was filled with 0 to represent no accumulated charge at the start of service.

6.2 Duplicate Records

No duplicate rows were found in the raw dataset (duplicate count = 0). Therefore, no observations were removed for duplication.

6.3 Data Types and Inconsistent Values

TotalCharges was converted from object/text to numeric. Other categorical labels were inspected and were consistent in spelling and capitalization. Churn was mapped to a binary modelling target only after the raw categorical values were preserved for interpretation.

6.4 Outlier Handling

The IQR method was used to inspect tenure, MonthlyCharges and TotalCharges. No observations fell outside the $1.5 \times \text{IQR}$ fences for these variables, so no numeric observations were removed as outliers.

Variable	Q1	Q3	IQR	Lower Fence	Upper Fence	Outliers
tenure	9.00	55.00	46.00	-60.00	124.00	0
MonthlyCharges	35.50	89.85	54.35	-46.02	171.38	0
TotalCharges	398.55	3786.60	3388.05	-4683.52	8868.67	0

6.5 Before vs After Cleaning

Quality Item	Before	After	Treatment
TotalCharges type	object	float	Numeric conversion
Blank TotalCharges	11	0	Filled with 0
Duplicate rows	0	0	No removal required
Rows	7043	7043	No row loss

7. DATA PREPROCESSING

7.1 Categorical Encoding

Categorical predictors are one-hot encoded with `handle_unknown='ignore'`, allowing unseen test categories to be handled safely.

7.2 Feature Scaling and Transformation

Numerical predictors are median-imputed and standardized for the Logistic Regression baseline. The same preprocessing pipeline is used for Random Forest for a consistent comparison; scaling does not drive tree splits but keeps the workflow uniform.

7.3 Train-Test Split

The dataset is divided into 80% training and 20% testing data using `random_state=42` and stratification by the churn target. Preprocessing is fitted inside the pipeline on training data only, preventing information leakage from the test set.

7.4 Final Feature Set

The final predictors include demographic, service, contract, billing and payment variables. `customerID` is excluded because it is an identifier without a meaningful behavioural interpretation. Churn is the target.

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

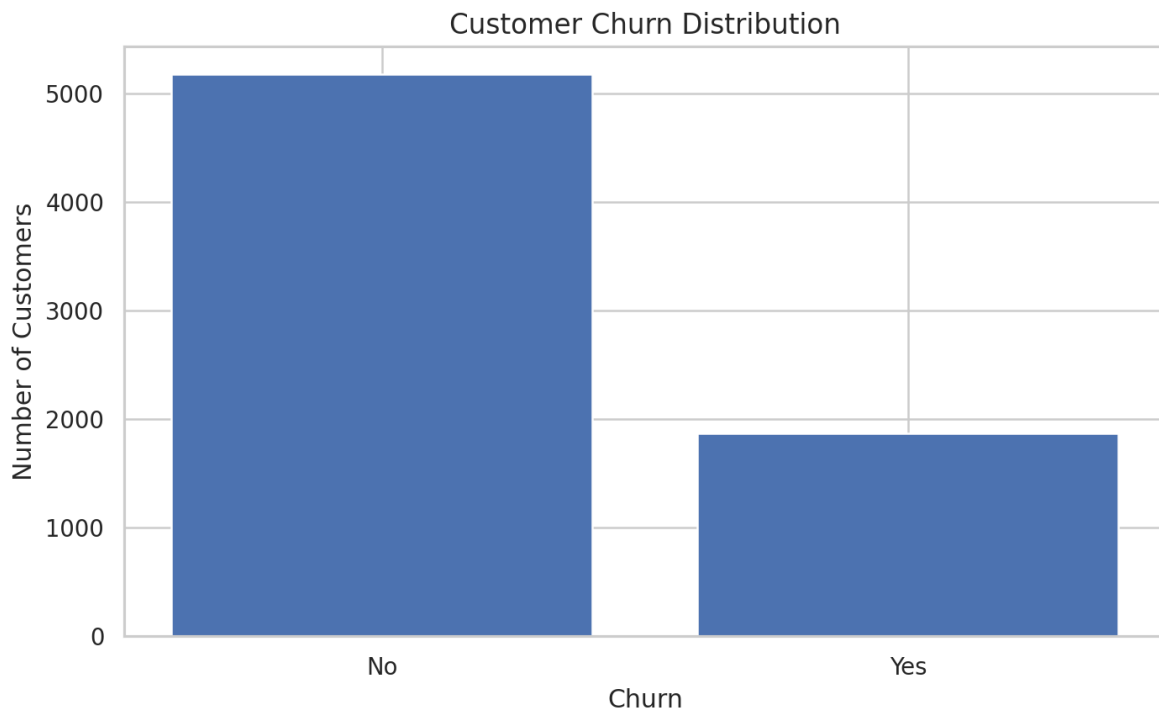


Figure 1. Distribution of customer churn outcomes.

Churn represents 26.54% of the sample, while 73.46% of customers are retained. This class imbalance makes recall, precision and F1-score important alongside accuracy.

8.2 Categorical Analysis

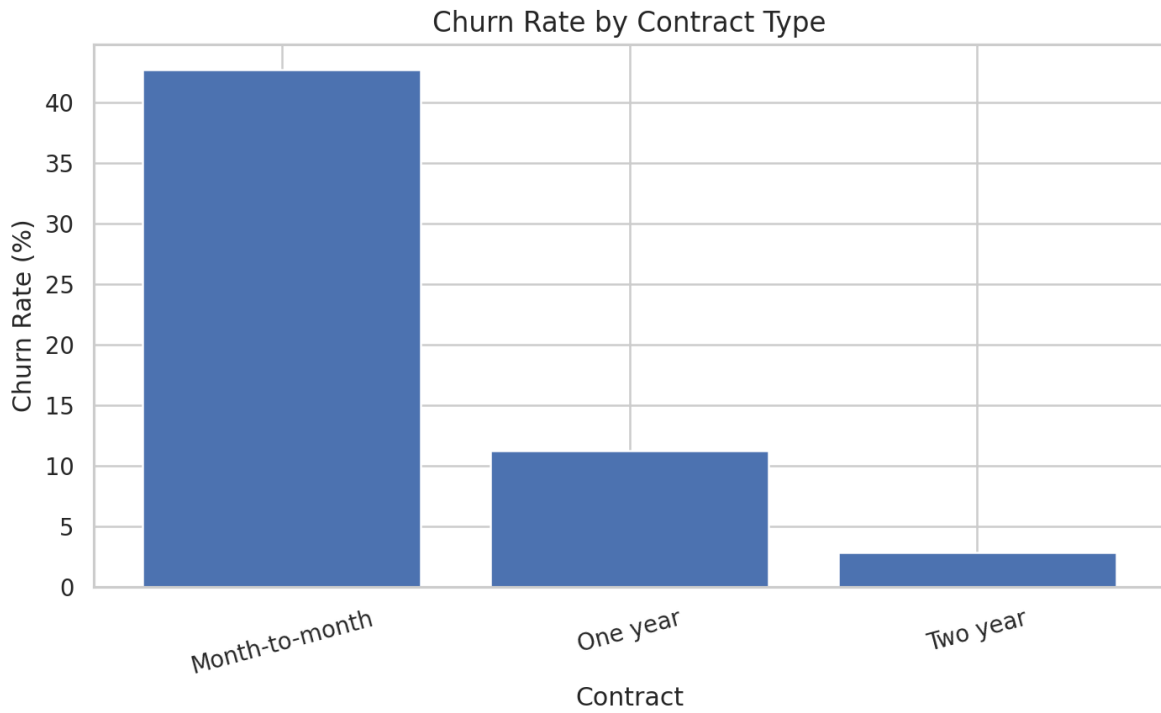


Figure 2. Observed churn rate across contract types.

Month-to-month customers show an observed churn rate of 42.71%, compared with 11.27% for one-year contracts and 2.83% for two-year contracts. Contract type is therefore a major segmentation variable.

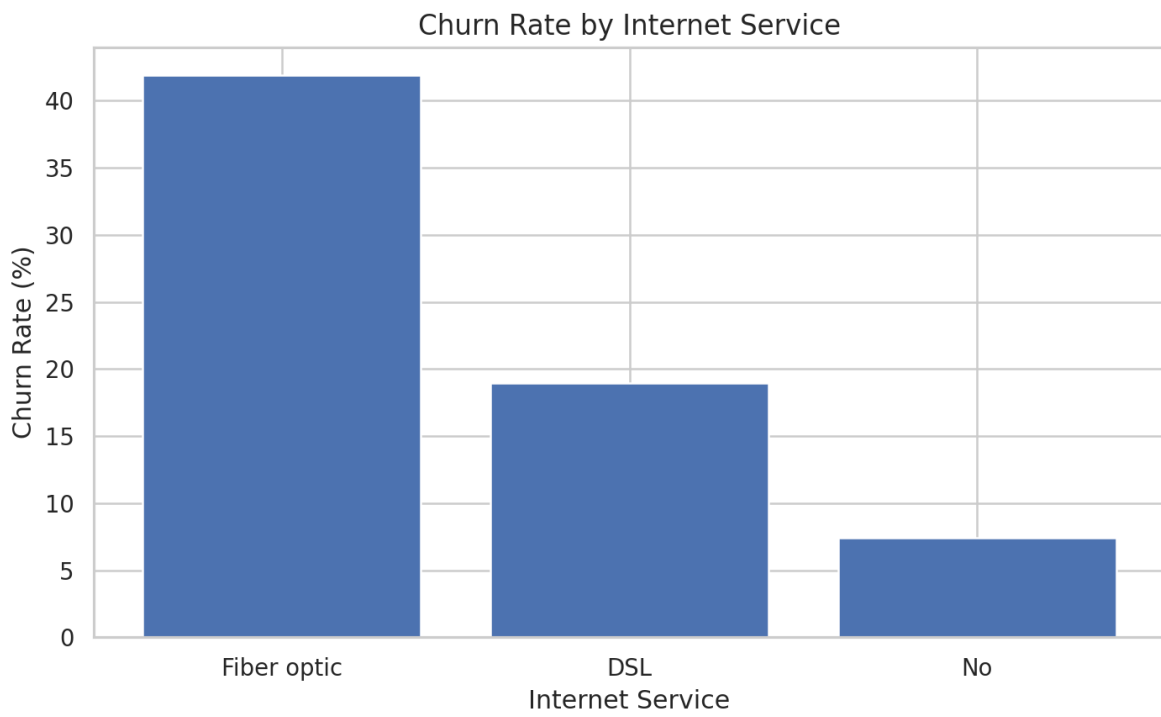


Figure 3. Observed churn rate across internet service types.

Fiber optic customers have an observed churn rate of 41.89%, while DSL customers show 18.96% and customers without internet service show 7.40%.

8.3 Bivariate / Relationship Analysis

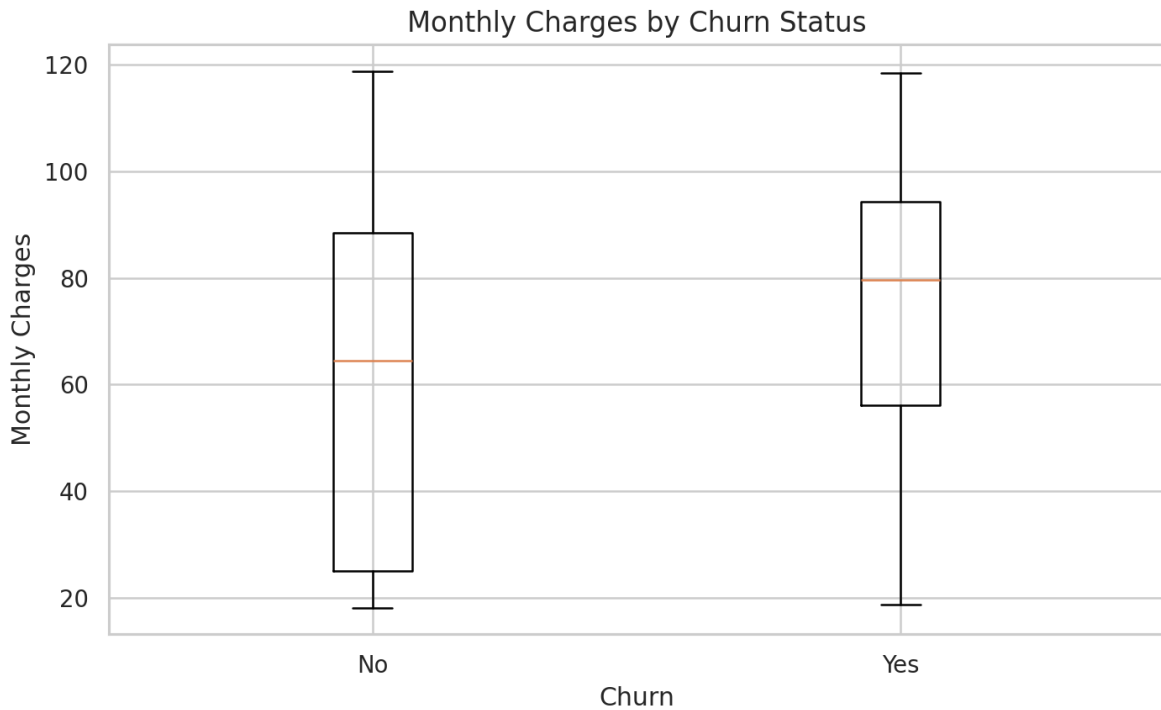


Figure 4. Monthly charges by churn outcome.

Average monthly charges are 74.44 for churned customers versus 61.27 for retained customers. This is an observed association and should not be interpreted as proof that charges cause churn.

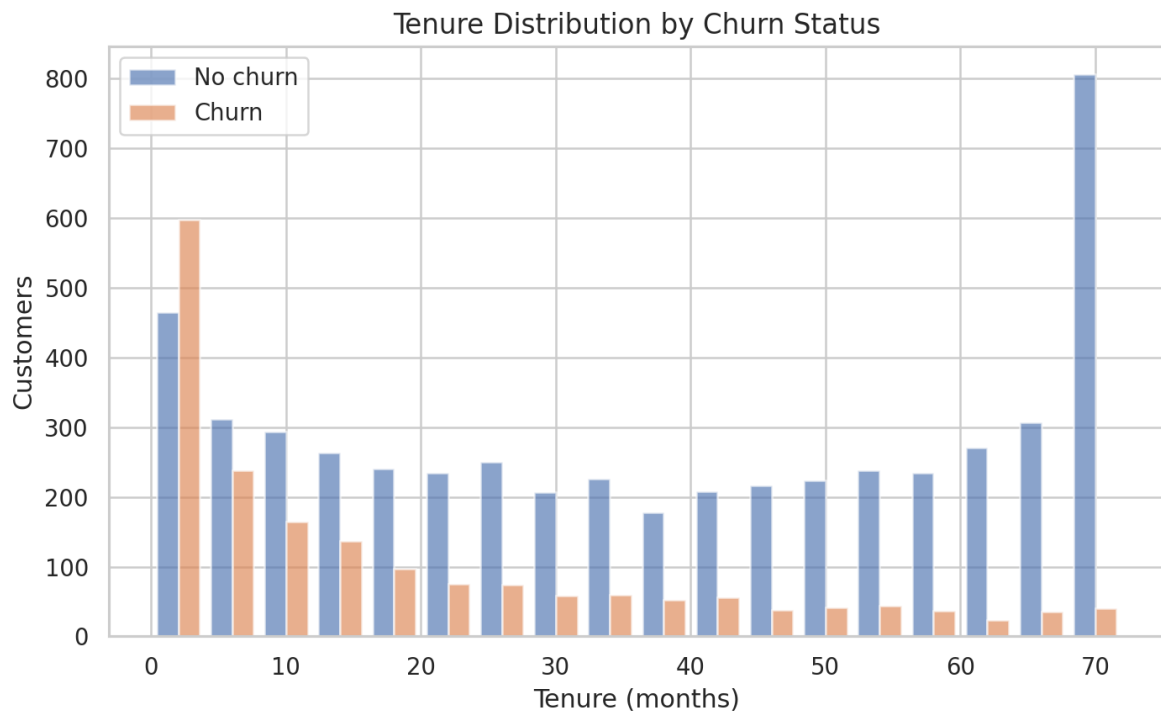


Figure 5. Tenure distributions by churn outcome.

Average tenure is 17.98 months for churned customers and 37.57 months for retained customers, showing a substantial lifecycle difference.

8.4 Correlation Analysis

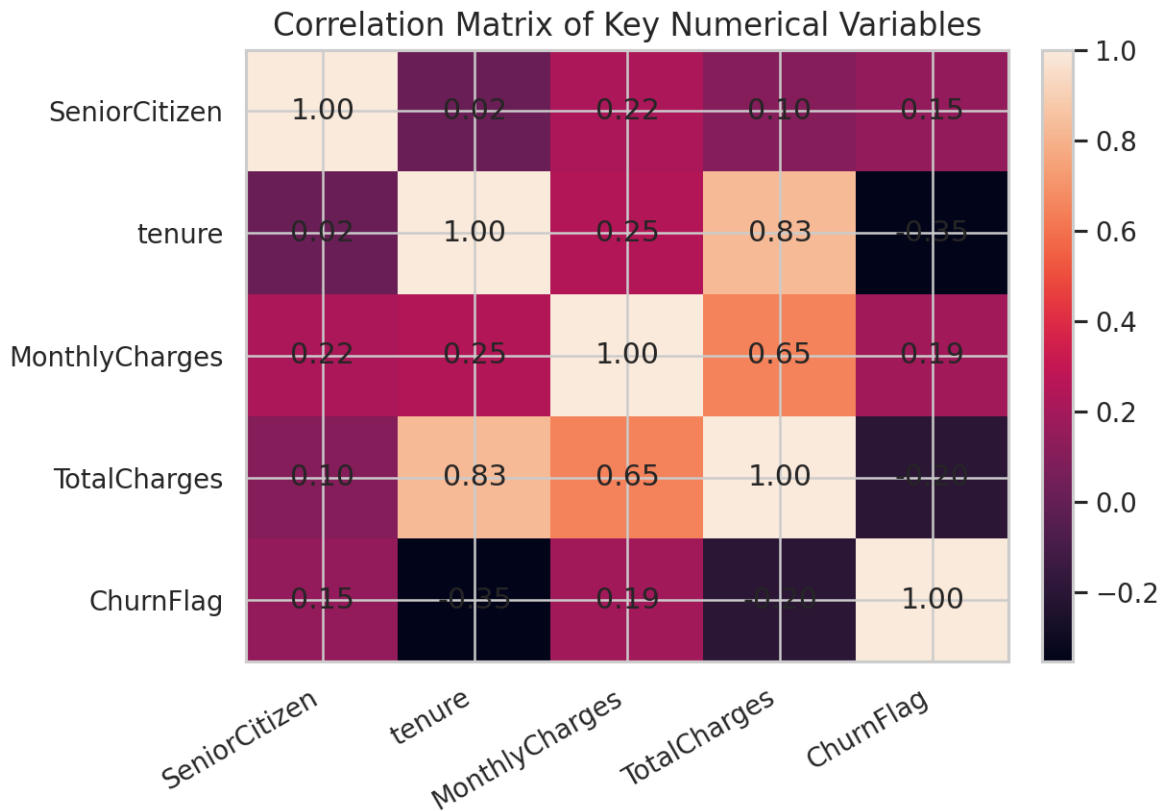


Figure 6. Correlation matrix of selected numerical variables and the churn indicator.

Tenure and TotalCharges are strongly related because accumulated charges grow with time in service. Churn has a negative relationship with tenure and positive relationships with several billing-related variables. Correlation supports feature understanding but does not establish causation.

8.5 Other Figures

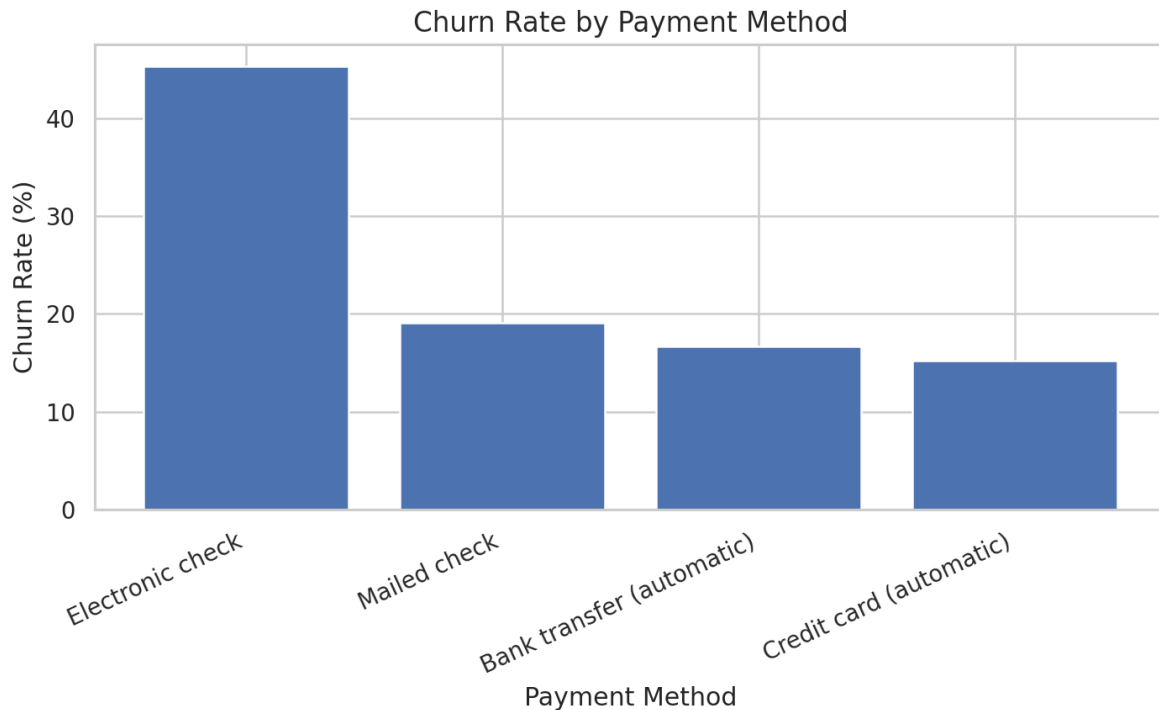


Figure 7. Observed churn rate by payment method.

Electronic check has the highest observed churn rate among payment methods at 45.29%. This may be useful for retention segmentation and payment-experience investigation.

9. STATISTICAL ANALYSIS

Variable	Mean	Median	Std Dev	Min	Max
Tenure	32.37	29.00	24.56	0	72
MonthlyCharges	64.76	70.35	30.09	18.25	118.75
TotalCharges	2279.73	1394.55	2266.79	0.00	8684.80

Group-wise statistics show that churned customers tend to have shorter tenure and higher monthly charges. The distribution of TotalCharges is naturally linked to tenure, so it should not be interpreted independently of service duration.

10. FEATURE ENGINEERING

The main engineered modelling feature is the binary ChurnFlag target (No=0, Yes=1). No artificial composite features were introduced because the original variables already capture contract, service and billing dimensions. Keeping the feature set interpretable supports the capstone's research-style emphasis.

11. MODEL DEVELOPMENT

11.1 Problem Formulation

This is a binary classification problem. The target is Churn, and predictors are customer demographics, tenure, service subscriptions, contract, billing and payment variables.

11.2 Model Selection

Logistic Regression is used as an interpretable baseline. Random Forest is used as the final nonlinear model because churn relationships may involve interactions among contract, tenure, services and billing characteristics.

11.3 Training the Model

Both models use the same leakage-safe preprocessing pipeline and an 80/20 stratified split. Random Forest uses 300 trees, maximum depth 8, minimum leaf size 3, class balancing and random_state 42.

12. MODEL EVALUATION

12.1 Classification Evaluation

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.806	0.657	0.559	0.604	0.842
Random Forest	0.763	0.537	0.783	0.637	0.842

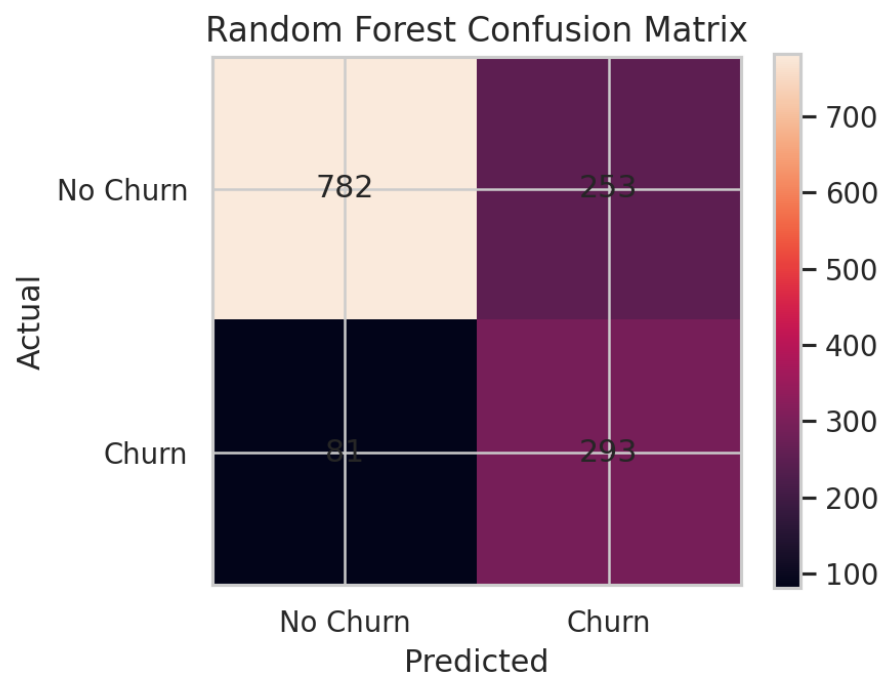


Figure 8. Random Forest confusion matrix on the 20% test set.

The Random Forest correctly identifies 293 churners and misses 81 churners in the test set. Its recall of 78.34% indicates that it identifies a larger share of actual churners than the Logistic Regression baseline.

12.2 Overfitting and Underfitting

Random Forest training accuracy is 0.797, compared with test accuracy of 0.763. The gap is 0.034, suggesting no severe accuracy overfitting under this configuration. Evaluation on unseen test data remains the primary evidence of generalization.

13. RESULTS AND FINDINGS

1. Overall churn rate is 26.54%, making churn a meaningful but minority outcome.

2. Month-to-month customers have the highest observed churn rate (42.71%), far above one-year and two-year contracts.
3. Churned customers have shorter average tenure (17.98 months) than retained customers (37.57 months).
4. Average monthly charges are higher among churned customers (74.44) than retained customers (61.27).
5. The Random Forest achieves 76.30% accuracy and 63.70% F1-score, with recall of 78.34%, making it a useful baseline for churn-risk screening.

14. DISCUSSION

The EDA and model results point to customer lifecycle and relationship structure as central themes. Contract commitment is strongly associated with observed churn, while shorter tenure marks a higher-risk stage. Higher monthly charges and particular service/payment configurations also distinguish churn outcomes. The Random Forest captures nonlinear patterns and provides higher recall and F1-score than the logistic baseline, which is valuable when the business wants to avoid overlooking potential churners. These findings are predictive associations, not causal effects.

15. LIMITATIONS

- The dataset is observational and cannot establish causality.
- Important churn drivers such as competitor offers, satisfaction changes, service incidents or customer complaints may be absent.
- Historical model performance may not generalize unchanged to future customer populations.
- Random Forest feature importance indicates predictive contribution rather than causal importance.
- Business-specific misclassification costs were not supplied, so the classification threshold was not optimized for a particular financial objective.

16. CONCLUSION

This capstone completed an end-to-end churn analysis on 7,043 telecommunications customer records. The analysis found substantial differences across contract types, tenure, monthly charges, internet service and payment methods. A Random Forest classifier provided the strongest balance of recall and F1-score among the tested models and can serve as a practical screening baseline. The evidence supports targeted retention, particularly for month-to-month and early-tenure customers, while emphasizing human review and ongoing model validation.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

- Prioritize proactive retention outreach for month-to-month customers because this segment has the highest observed churn rate.
- Strengthen onboarding and early-lifecycle engagement because churned customers have substantially shorter tenure.
- Review pricing and value communication for higher-charge customers, while testing retention offers rather than assuming price is causal.

- Investigate payment-method experience, especially electronic-check customers, where observed churn is highest.
- Use the Random Forest as a risk-screening tool and combine its predictions with customer-service context before taking action.

17.2 Future Work

Future work could include probability-threshold tuning based on the cost of missed churners versus unnecessary outreach, temporal validation, calibrated probabilities, additional customer-interaction variables, and controlled retention experiments to measure whether interventions actually reduce churn.

18. REFERENCES

India Data Research Academy. (2026). Capstone Project Guide 2026. Data Science & AI Training Program.

India Data Research Academy. (2026). Capstone Project Template Sample. Data Science & AI Training Program.

Pandas documentation. Data manipulation and analysis in Python.

Scikit-learn documentation. Classification, preprocessing, pipelines and model evaluation.

APPENDIX A. CODE

The complete reproducible code is provided in `'Customer_Churn_Capstone_Analysis.ipynb'`. It contains data loading, cleaning, IQR analysis, EDA, statistical analysis, preprocessing, model training, evaluation and interpretation.

A.1 Data Loading and Inspection

The notebook loads the LMS CSV, displays the first rows, shape, data types, missing values and duplicates.

A.2 Data Cleaning

TotalCharges is converted to numeric and its 11 blank values are resolved using the documented tenure-based treatment.

A.3 Preprocessing

Categorical one-hot encoding, numerical imputation/scaling and the stratified train-test split are implemented using a scikit-learn Pipeline.

A.4 EDA and Visualization

The notebook generates churn distribution, contract, internet-service, monthly-charge, tenure, payment-method, correlation and feature-importance figures.

A.5 Model Training and Evaluation

Logistic Regression and Random Forest are trained and compared using accuracy, precision, recall, F1-score and ROC-AUC, with a confusion matrix for the final model.

APPENDIX B. ADDITIONAL FIGURES AND TABLES

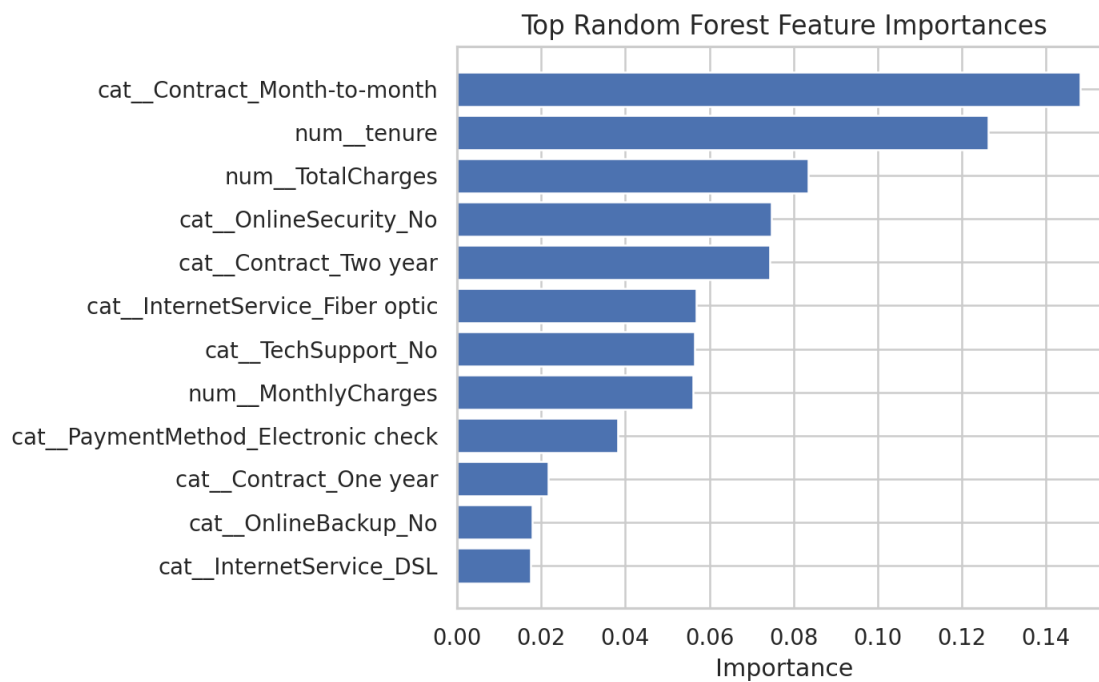


Figure 9. Top Random Forest feature importances.

The feature-importance chart highlights contract, tenure, security/support, internet service, payment method and billing variables among the strongest predictive inputs.

APPENDIX A — KEY CODE EXCERPTS (REPRODUCIBILITY)

The complete executable code remains in the Jupyter notebook. The excerpts below document the most important analytical and modelling decisions, consistent with the IDRA guidance.

A.1 Data Loading and Inspection

```
# Data loading and inspection
df = pd.read_csv("P_4_WA_Fn-UseC_-Telco-Customer-Churn.csv")
print(df.shape)
display(df.head())
display(df.dtypes.to_frame("dtype"))
```

A.2 Data Cleaning

```
# Cleaning TotalCharges
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
print(df["TotalCharges"].isna().sum()) # 11 blank records
df["TotalCharges"] = df["TotalCharges"].fillna(0)
```

A.3 Preprocessing

```
# Leakage-safe preprocessing and split
X = df.drop(columns=["Churn", "customerID", "ChurnFlag"])
y = df["ChurnFlag"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
```

A.4 Model Training and Evaluation

```
# Model training and evaluation
pipe = Pipeline([("preprocessor", preprocessor), ("model", model)])
pipe.fit(X_train, y_train)
pred = pipe.predict(X_test)
prob = pipe.predict_proba(X_test)[:, 1]

f1 = f1_score(y_test, pred)
recall = recall_score(y_test, pred)
roc_auc = roc_auc_score(y_test, prob)
```

A.5 Classification Evaluation

```
# Confusion matrix
cm = confusion_matrix(y_test, pred)
print(cm)
```

REFERENCE NOTE

Primary dataset: India Data Research Academy (IDRA), Project 4 — Telco Customer Churn dataset, supplied through the LMS Study Material.

Methodology references: pandas documentation; scikit-learn documentation for preprocessing pipelines, classification models and evaluation metrics.

The report intentionally does not cite external churn studies for empirical claims that were calculated directly from the supplied dataset.